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Cognitive requirements for tool use by New Caledonian crows (*Corvus moneduloides*)

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Abstract Cumulative technological evolution has been suggested to explain the existence of different pandanus tool designs manufactured by New Caledonian crows. Circumstantial evidence from the distribution of the three tool designs that they manufacture suggests transmission of the designs probably involves accurate social learning, a characteristic considered essential for the cumulative evolution of tools. Recently, Kenward et al. (2005) reported that four hand-raised crows developed basic stick tool use without social learning. This finding cast doubt on the importance of social learning in the evolution of crows' pandanus tools in the wild. Here, we report that a naïve male crow at Parc Zoo-Forestier, Nouméa, developed proficient stick tool use without social input in 2002. In 2004, four captive crows, including the naïve male, that were inexperienced with pandanus material were given an opportunity to use and/or manufacture pandanus tools. Only two of the four birds used the tools but none manufactured tools. Our preliminary findings and the work with the four hand-raised crows keep open the possibility that the evolution of crows' pandanus tool designs is based on social learning. We propose that social learning and a disposition to develop basic tool use without social input are both essential cognitive requirements for cumulative technological evolution.

Keywords *Corvus moneduloides*; cumulative technological evolution; New Caledonian crows; ontogeny; social learning; tool manufacture and use

INTRODUCTION

A crucial development enabling the evolution of sophisticated technologies is the ability to improve tools in a cumulative way (Tomasello et al. 1993). This cumulative process requires the accurate social transmission of technical knowledge across generations, with each generation having the opportunity to make technological advances before the knowledge is passed on once again. A simple example in humans is the gradual increase in sophistication of the axe. Theory relating to cumulative technological evolution (CTE) is predominantly based on what we know about the cognition of modern humans (with relatively advanced CTE) and non-human primates (without CTE). The lack of even rudimentary CTE in non-human animals has meant researchers often focus on trying to identify what is "unique" about human cognition (e.g., Tomasello et al. 1993). The existence of human and non-human models of CTE would be extremely valuable for identifying cognitive characteristics common to this level of technology.

New Caledonian crows (*Corvus moneduloides*) (crows hereafter) manufacture tools more complex than other non-humans (Hunt 1996). The only credible evidence for rudimentary CTE has come from these crows (Hunt & Gray 2003). In the wild, they manufacture three specific tool designs from the leaves of *Pandanus* trees to extract food: wide, narrow and stepped designs (Fig. 1) (Hunt & Gray 2003). The stepped design is the most complex shape and its manufacture requires the longest sequence of

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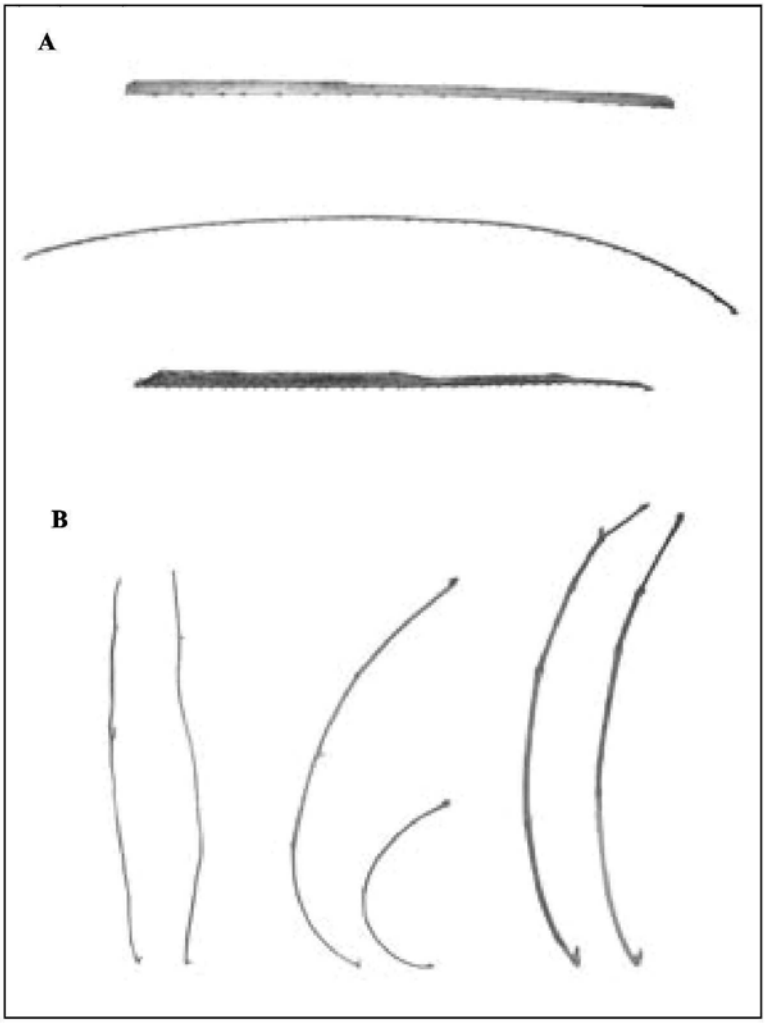


Fig. 1 Examples of barbed-pandanus tools and hooked-stick tools manufactured by New Caledonian crows. **A**, A wide (top), narrow (middle) and two-stepped pandanus tool (bottom). The length of the wide tool is 18.4 cm. **B**, Hooked tools made from fern stolons (two tools at left), a thorny vine (two middle tools) and forked twigs (two tools at right). The length of the fern stolon at far left is 13.9 cm.

behavioural manipulations. Hunt & Gray (2003) presented evidence that suggested the three pandanus designs have the same evolutionary lineage and that the stepped design evolved from incremental changes to simpler designs like the wide design. Recent support for the idea that stepped tools are the most complex design is that their manufacture appears to require greater functional specialisation of the brain than the manufacture of wide or narrow tools (Hunt et al. 2006). Hunt & Gray (2003) suggested that the transmission of the three designs probably involves social learning. They based this on the following circumstantial evidence: (i) consistency in the variety of design made at different sites; (ii) consistency in the shape of a particular design at individual sites, but shape variation between sites; (iii) the unique geographic distributions of each design that are more or less continuous; and (iv) that the distributions also overlap in a way that suggests the designs spread out from a "centre of innovation" in the south-east of the mainland. That the distributions of the three designs are continuous and overlapping indicates that design variation is unlikely to be an adaptive response to different environmental conditions. The absence of social learning in the transmission of crows' complex tool manufacture could have implications for what prerequisites are required for rudimentary CTE to evolve.

Direct evidence for how tool skills develop in crows comes from work with four hand-raised

juveniles (Kenward et al. 2005, 2006). The authors raised the four crows from an early age in artificial nests. Two of the juveniles were housed together in the same cage and the other two were housed alone. The juveniles were provided with sticks and food that could be extracted with tools. All four crows first successfully extracted food with sticks at around 70 days old. Several weeks after they developed stick tool use, the crows were presented with pandanus leaves. They all ripped the material with their bills, but only one bird manufactured and used "rough" pandanus tools to extract food. That only one of the four crows developed this behaviour when first presented with pandanus leaves raises the possibility that pandanus tool skills develop differently to stick tool skills. Although Kenward et al. (2005) cautioned that crows' more complex manufacture techniques in the wild might require social learning, as Hunt & Gray (2003) had suggested, a clear implication from their findings is that basic pandanus tool skills, as well as basic stick tool skills, develop from an inherited disposition. We think that it is unwise to draw conclusions about the ontogeny of basic pandanus tool skills based on positive evidence from only one out of four individuals.

Here, we describe how in 2002 a naïve male at Parc Zoo-Forestier, Nouméa, developed stick tool use when isolated from other crows, an observation consistent with the findings of Kenward et al. (2005). Two years later we conducted a simple, brief experiment at the zoo with this male and three wild-caught adult crows also held at the zoo to see how they responded to pandanus leaves and tools that we provided for them. The preliminary results provide further support for our suggestion that crows' basic pandanus tool skills develop differently from their basic stick tool skills, and lead to a novel proposal concerning essential prerequisites for CTE based on the development of tool skills in crows.

DEVELOPMENT OF STICK TOOL USE IN A NAÏVE CROW

A male close to fledging was rescued at Uitoe, Grande-Terre, after it had fallen from its nest (crow 1 in Table 1). After arriving at Parc Zoo-Forestier, Nouméa, on 1 December 2001 he was initially housed alone. The male was hand-fed for 3 days until he independently took food in a dish, which he has continued to be provided with each day. In early 2002, an experiment was carried out to see if he could develop stick tool use without having had contact with other crows. This work was conducted more out of curiosity and not documented.

At first, sticks were scattered on the floor of the male's cage. On several occasions, his keepers placed sticks in his bill to see his reaction to them. At the beginning of April 2002, the male also had access to food in vertical holes that could be extracted with tools. His keepers never demonstrated tool use. The male manipulated the sticks placed in his bill and those on the floor of his cage in two main ways: he passed them through the mesh walls or used them to poke at the door latch. It took about 1 week for the male to use sticks to try and extract food in the holes. After 2 weeks he was very efficient at using sticks to extract the meat. The development of proficient stick tool use by the naïve male at Parc Zoo-Forestier provides independent support for the behaviour of four hand-raised crows described by Kenward et al. (2005). That is, stick tool use in crows can develop without learning from conspecifics, probably through a disposition for directed object manipulation combined with trial and error. Since 22 October 2002, a wild-caught adult female crow shared the male's cage (crow 2 in Table 1).

HOW DOES PANDANUS TOOL MANUFACTURE AND USE DEVELOP?

In June 2004, we carried out a simple experiment with four crows housed at Parc Zoo-Forestier. The crows were the naïve male described above and three wild-caught birds netted as adults in farmland at Boulouparis, Grande-Terre; one of the three wild-caught crows (an adult female) shared the naïve male's cage and the other two were an adult pair caged together out of sight of the other two crows (crows 3 and 4 in Table 1). It is likely that none of the four crows had any experience with pandanus trees in the wild, which are rare in the areas where the birds were captured. The four crows had no experience with pandanus material following capture.

Table 1 Study crows.

Crow	Sex	Age	Origin
1	male	<3 years	captured just before fledging in December 2001
2	female	>3 years	wild-caught in mid 2001
3	female	>3 years	wild-caught in mid 2001
4	male	>3 years	wild-caught in mid 2001

Given their probable lack of experience with pandanus trees, we took the opportunity to investigate if they could spontaneously manufacture and/or use pandanus tools. The juvenile crow that made a pandanus tool as described by Kenward et al. (2005) did so on the first day it was given leaf material. We conducted the experiment in the two aviaries where the crows were housed.

For the first stage of the experiment, we provided the four crows with meat visible at the base of vertical holes 6–7 cm deep × 2 cm diameter in a dead log, along with sticks on the floor of their cage to use as tools. We also provided eight pandanus leaves tied in a fan arrangement to a dead branch, which we positioned next to the log with the meat-baited holes. This set up was used in both aviaries for 2 days. For the second stage on days 3 and 4, we removed all potential stick tools and supplied four 12 cm long wide-tool-like strips of barbed pandanus leaf-edge placed in meat-baited holes. In each trial, we placed a pandanus tool in four holes. The leaf-edge barbs faced upwards, allowing a tool to be immediately used to hook out meat. We conducted three trials with the female and naïve male, and seven with the pair. A trial finished when either all meat had been removed from the holes or c. 30 min had elapsed. The number of trials that we conducted was small because we were only interested in crows' initial reaction to the pandanus material. Corvids, especially older individuals, can be very wary of novelty and avoid it (Heinrich 1995). However, our aim was to test crows' initial reaction to pandanus material that they probably had no previous experience with. If crows have an inherited disposition to use this material for tools, we expected them to do so without hesitation.

Table 2 Summary of pandanus tool use experiment.

Crow	Trial	Behaviour
Cage one		
1	1	tool use but no meat extracted
1	2	1 piece of meat extracted (possibly by chance when tool was removed from the hole with meat snagged onto the end of it)
1	3	2 pieces of meat extracted after tool use
2	1	did not interact with pandanus tools
2	2	did not interact with pandanus tools
2	3	did not interact with pandanus tools
Cage two		
3	1	tool use but no meat extracted
3	2	1 piece of meat extracted by chance 1 piece of meat extracted after tool use
3	3	tool use but no meat extracted
3	4	discarded pandanus tools without using them
3	5	1 piece of meat extracted by chance
3	6	did not interact with pandanus tools
3	7	did not interact with pandanus tools
4	1	did not interact with pandanus tools
4	2	did not interact with pandanus tools
4	3	did not interact with pandanus tools
4	4	did not interact with pandanus tools
4	5	did not interact with pandanus tools
4	6	discarded pandanus tools without using them
4	7	1 piece of meat extracted by chance

During stage one of the experiment, all four crows extracted meat efficiently from the holes with stick tools but only the naïve male manipulated the pandanus leaves. He tore the leaves indiscriminately into shapes which did not resemble wide, narrow or stepped tools, and sometimes removed sections of leaf material. We never saw him use the torn pieces as tools. In stage two, only the naïve male and the paired female (crow 3) demonstrated definite use of the provided pandanus tools to extract meat (Table 2). The female caged with the naïve male never used the pandanus tools and the paired male (crow 4) extracted one piece of meat accidentally when he pulled out the tool to discard it (Table 2). Although the naïve male and the paired female extracted meat after tool use, we do not know if the meat was actively hooked out, as experienced crows are thought to do (Hunt & Gray 2004a), or became passively snagged onto the end of the tool. After we had tried to remove all potential stick tools from the cage of the crow pair in stage two, the pair sometimes made stick-like tools out of fern and grass stems that they found in the cage, but never from the pandanus leaves that we provided. That they found stick-like material in their cage could have affected their incentive to use the pandanus material for tools.

COGNITIVE PREREQUISITES FOR CUMULATIVE TECHNOLOGY

It is not surprising that an inherited disposition evolved for tool use in crows that is strongly associated with sticks. Play in *Corvus* species often involves sticks (e.g., Kilham 1989) and they use sticks to build their nests. Sticks are the commonest form of tool material used by animals (Beck 1980) and the commonest raw material for tools in New Caledonian forests. Stick tool use is also probably

more widespread than pandanus tool use over the crows' total range because pandanus trees have a patchy distribution (G. Hunt pers. obs.).

On the other hand, there are reasons to suspect that pandanus tool skills develop differently to basic stick tool skills. First, in contrast to the use of sticks, there is no phylogenetic history in other *Corvus* species for the use of pandanus leaf in any way and crows use the material only for tools (Hunt 2000). Second, the infrequent display of pandanus tool skills in the eight captive crows described here and by Kenward et al. (2005) contrasted strikingly with proficient stick tool use by all of them. Third, an experimental effect of social learning was demonstrated in the hand-raised juveniles that Kenward et al. (2005) tested. Kenward et al. (2006) documented the ontogeny of stick tool use in the four crows under different conditions; two of the four crows received demonstrations of tool use and the other two did not. Rates of tool handling and tool use by the former two crows were higher than that for the crows which were not tutored. Last, the pieces of pandanus material that the naïve zoo male and the four crows described by Kenward et al. (2005) tore off the provided leaves did not resemble the shapes of either the wide, narrow or stepped tool designs. This demonstrates that individuals do not spontaneously manufacture any of the three known designs through some kind of "genetic-based disposition".

The four crows described by Kenward et al. (2005) and the naïve male at the zoo all indiscriminately ripped pandanus leaves put into their cages. That the adult pair at the zoo and the female housed with the naïve male did not touch the pandanus leaves suggests the ripping by naïve birds might come from a youthful curiosity to explore and pull apart new material. Curiosity and object-directed play are common in juveniles of other *Corvus* species (Kilham 1989; Heinrich 1995). Kenward et al. (2005) provided pandanus leaves (after removing all stick material) to the four hand-raised juveniles only after they were proficient at using stick tools. Given that only one of the four juveniles made or used pandanus tools, it is possible that its behaviour was an innovation based on trial and error learning. That is, its previous experience at extracting food with sticks and ripping pandanus leaves helped it to make tools from the only potential material available. The behaviour of the naïve male at the zoo also suggests that pandanus tool manufacture by crows raised in isolation and experienced with stick tools might develop only in resourceful individuals. The adult pair at the zoo was keen to extract food with sticks but not with strips of pandanus leaf. New tool skills might generally only develop in a specific "learning window" when crows are very young, as in tool-using woodpecker finches *Cactospiza pallida* (Tebbich et al. 2001). Kenward et al. (2005) provided pandanus material approximately 30 days after the hand-raised juveniles first successfully used stick tools to extract food, therefore it was still likely to be within any learning window for tool use.

Pandanus tool manufacture in the wild is highly "channelled" into producing a particular shape of pandanus tool at a particular site. There is no evidence from captive experiments that these population-level designs can be produced by naïve crows. Trial and error learning or shapes highly constrained by local environmental conditions are also unlikely to account for the geographic distribution patterns of the designs (Hunt & Gray 2003). The geographic distribution of the three tool designs and the absence of pandanus tool manufacture by four of the five naïve captive crows together raise the possibility that the manufacture of each pandanus tool design involves social learning, as Hunt & Gray (2003) have suggested. Parsimony suggests that an inherited disposition to develop basic stick tool skills evolved early on in the history of crows' tool behaviour. After this disposition was in place, we propose that (i) pandanus tool technology evolved through individual and social learning, and (ii) crows' basic stick tool skills were enhanced through learning to include the manufacture of hooked tools (Hunt & Gray 2002, 2004b; examples in Fig. 1).

If the transmission of pandanus tool manufacture involves social learning, what is the mechanism(s) likely to be? Mechanisms that allow the social learning of information can range in cognitive sophistication from low-level stimulus enhancement to high-level imitation. True imitation has been defined as "...a cognitively sophisticated task, calling for a quite abstract, functional representation of a problem and its solution" (Sterelny 2003). True imitation is considered to be essential for CTE because it is thought to be the only way in which complex techniques can be transmitted incrementally across generations (Tomasello et al. 1993; Tomasello 1999; Sterelny 2003). There is little evidence for true imitation in wild animals (Laland & Hoppit 2005), including crows, which would seem to be at odds with the claim of CTE in crows' pandanus tool technology. However, the CTE that Tomasello et al. (1993) and Sterelny (2003) have in mind ratchets up at a fast pace and developed only very recently in human evolution (<100 000 years BP). Our definition of

CTE is much broader to include evidence for this technology even when the ratchet seems to be turning slowly. For example, stone tool technology does show incremental increases in complexity during the lower paleolithic, albeit at a slow rate. The over 1-million-year-long Acheulean technological period characterised by the so-called "handaxes" is generally accepted as an improvement to the earlier Oldowan stone tool technology (Donald 1991). We agree with Sterelny (2003) that slow-changing stone tool traditions like the Oldowan and Acheulean were probably associated with less sophisticated cognitive social learning mechanisms than are required for true imitation, and heavily reliant on trial and error learning. This view is supported by recent evidence obtained by imaging brains of humans making Oldowan-type stone tools (Stout & Chaminade 2006). The brain regions that were activated were "lower-level" ones responsible for sensorimotor capabilities, not "higher-level" regions in the prefrontal cortex responsible for abstract conceptualisation and planning. Traditions may be based on many kinds of social learning (Avital & Jablonka 2000) and Laland & Hoppit (2003) argue that there is no evidence imitation is necessary for cumulative culture. We do not expect crows' pandanus tool technology to be based on true imitation. However, as we explain above, we do not think that the absence of true imitation invalidates the claim of CTE in crows' tool technology, as long as the CTE is based on a cognitively less sophisticated social learning mechanism.

That basic stick tool skills can develop from an inherited disposition has raised questions about the degree to which socially learned behaviours reported in non-humans are inherited genetically (Laland & Janik 2006). However, the expression of a complex behaviour may require both social learning and an inherited disposition for the basic skills. For example, although modern human technology is obviously the result of cumulative cultural evolution, humans probably also have a predisposition for basic tool use. The development of human tool use is thought to begin when infants are less than 12 months old and before they have developed skills like relational reasoning and insight with perception-action routines (Lockman 2000). Similarly, humans probably have an inherited disposition for acquiring a language, but aspects of language such as a specific lexicon are obviously socially learnt (Pinker 1994). In a process possibly analogous to language evolution, a disposition to develop basic tool skills permits endemic tool use, which provides a "network" for the potential spread of design innovations if and when a species has the capacity to socially transmit them. Rather than being mutually exclusive, we propose that high fidelity social learning and a predisposition for basic tool use without social input are both essential cognitive requirements for cumulative technological evolution.

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